

ABSTRACT

Cardiovascular diseases remain the leading cause of death worldwide, accounting for approximately 17.9 million lives each year. However, mortalities associated with these diseases disproportionately affect the low-resource regions. One of the big challenges in fighting this crisis is the asymptomatic nature of early-stage heart disease and limited advanced diagnostic infrastructure in developing countries, often resulting in late diagnoses and deaths that could have been prevented. This study tackles the urgent need for accessible early detection tools by developing and evaluating a machine learning-based system to predict heart disease risk using standard clinical and demographic data.

Based on a data science and analytics approach, the present research used publicly available datasets for training and testing robust algorithms like Logistic Regression, Random Forest, and XGBoost. Much emphasis has been given in this work not only to ensuring strong predictive accuracy but also to ensure model interpretability by means of explainable AI techniques that support clinical trust and practical usefulness. The expected output of this research is a cost-effective, easily scalable diagnostic tool that will be useful for healthcare professionals in the early identification of high-risk patients with severe symptoms. Ultimately, this research aims to reduce premature mortality by 20-30% and further health equity, hence contributing to the United Nations' Sustainable Development Goal 3.

KEYWORDS

Keywords: Machine Learning, Heart Disease Prediction, Cardiovascular Disease (CVD), Design Science Research (DSR), Explainable AI (XAI), Predictive Modeling, Healthcare Equity.

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Finally, I acknowledge the custodians of the public datasets, specifically the UCI Machine Learning Repository and Kaggle, whose commitment to open data facilitated the empirical foundation of this study.

DEDICATION

This project is dedicated to Dr. Patrick Nabongo, a qualified medical doctor whose commitment to helping the sick and suffering inspired me to pursue a path that values human impact over the pursuit of profit. His example shaped my understanding of how knowledge and technology can be used in service of people. This work stands as a reflection of those influences and the values they represent. Special dedication goes to Linda Ochwada, for her constant encouragement, belief in my abilities, and quiet strength throughout my academic journey. I also dedicate this work to my nephew, Mansa Malo, whose curiosity, innocence, and joy at such a young age continue to remind me why the future is worth building with care and responsibility.

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Figure 3.1: Design Science Research (DSR) Process Cycle

LIST OF ABBREVIATIONS

AI – Artificial Intelligence

AUC – Area Under the Curve

CNN – Convolutional Neural Network

CVD – Cardiovascular Disease

DSR – Design Science Research

ECG – Electrocardiogram

EDA – Exploratory Data Analysis

HCI – Human-Computer Interaction

IPO – Input-Process-Output

LIME – Local Interpretable Model-Agnostic Explanations

LSTM – Long Short-Term Memory

ML – Machine Learning

SHAP – SHapley Additive exPlanations

UCI – University of California, Irvine (Machine Learning Repository)

WHO – World Health Organization

XAI – Explainable Artificial Intelligence

XGBoost – Extreme Gradient Boosting

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Heart disease is one of the main causes of death in the world today. According to the World Health Organization (2023), cardiovascular diseases kill about 17.9 million people each year, making up nearly one-third of all global deaths. These conditions often come from preventable causes such as poor diet, smoking, lack of exercise, and unmanaged stress. In many developing countries, the lack of affordable screening and qualified medical professionals makes the situation even worse. Tools used for diagnosing heart disease such as angiograms and ECGs often cost a lot or are not easy to get (Mensah, Roth, and Fuster, 2019). Because of this, researchers are looking for ways to use computer programs and large data collections to help detect risks early. Machine learning can look at large sets of patient data and find patterns that might not be obvious to doctors or through traditional analysis, predicting who is more likely to develop heart disease by studying variables such as cholesterol levels, blood pressure, and age. This study explores how machine learning can improve early heart disease prediction using available medical datasets, focusing on accuracy, interpretability, and real-world applicability.

1.2 Problem Statement

Despite global efforts, early screening tools remain expensive and inaccessible. Traditional methods such as electrocardiograms, echocardiography, and angiography require advanced equipment and professional expertise unavailable in most developing countries (Mensah, Roth, and Fuster, 2019). Machine learning can process existing patient information to identify at-risk individuals without invasive or costly tests. However, most published models have been tested on single datasets, limiting their reliability across different populations. This study addresses that gap by training and validating models on two independent datasets to create a tool that predicts accurately and works in real-world settings, contributing to SDG 3 (United Nations, 2022).

1.3 Research Objectives

1.3.1 General Objective

To develop and evaluate machine learning models that accurately predict heart disease using clinical and demographic data, and to deploy the resulting system as an accessible web-based artifact.

1.3.2 Specific Objectives

1. To design and evaluate Logistic Regression, Random Forest, and XGBoost models for heart disease prediction.
2. To identify the most significant clinical variables influencing heart disease risk.
3. To compare model performance across two independent datasets to test generalizability.
4. To assess model interpretability and demographic fairness for clinical utility.

1.4 Rationale, Scope and Delimitations

This study is timely because millions suffer annually from preventable cardiac conditions while advanced diagnostic technologies remain out of reach in low-resource settings. Two open-access datasets are used: the UCI Heart Disease dataset (<https://archive.ics.uci.edu/dataset/45/heart+disease>) and the Kaggle Heart Failure Prediction dataset (<https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>). The study is limited to Logistic Regression, Random Forest, and XGBoost; deep learning is excluded due to resource and scope constraints. Findings are based on secondary data and may not capture all variations in lifestyle or genetic backgrounds. Further local validation is recommended before clinical deployment.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature on the application of machine learning in cardiovascular disease prediction. It begins by establishing the medical and social context of heart disease as a leading global health challenge requiring technological intervention. The review examines theoretical underpinnings of predictive analytics, evidence-based medicine, and socio-technical systems, before evaluating clinical symptom-based diagnosis against computational approaches. A critical appraisal of prior studies reveals significant algorithmic advances alongside persistent challenges in data quality and model interpretability. The chapter further addresses health resource inequity and the potential of predictive modelling to improve outcomes in underserved populations. By synthesising findings across sources, the chapter identifies key gaps in existing models that this study seeks to address, culminating in a clear research gap that informs the methodological approach of the succeeding chapter. Collectively, the literature confirms that while predictive analytics holds considerable promise, substantial work remains to make these models clinically trustworthy and broadly applicable.

2.2 Review of the Problem Domain

Heart disease remains the world's leading cause of death, with approximately 17.9 million lives lost annually, accounting for roughly 32% of all global deaths (World Health Organization, 2023). This crisis is especially severe in low- and middle-income countries where healthcare systems emphasize curative rather than preventive care. Changing lifestyle habits including sedentary behaviour and chronic stress are increasing the incidence of heart disease in younger age groups (Popkin, 2020). Conventional diagnostic methods are reactionary, addressing problems after symptoms have manifested significantly, which limits the window for effective intervention (Mensah, Roth, and Fuster, 2019). The inability of current systems to screen large populations at low cost represents a critical failure point in global health infrastructure.

2.3 Conceptual Framework

This study uses the Input-Process-Output (IPO) framework to outline how patient data are transformed into actionable clinical insights. The Input layer consists of patient demographics and clinical variables including age, blood pressure, cholesterol, and exercise-derived cardiac markers.

These enter the Process layer representing the machine learning pipeline: preprocessing, feature selection, and algorithmic processing with Logistic Regression, Random Forest, and XGBoost, enhanced with SHAP-based explainability. The Output layer consists of risk classification, probability score, and interpretability report designed for clinical decision support (Almustafa, 2020).

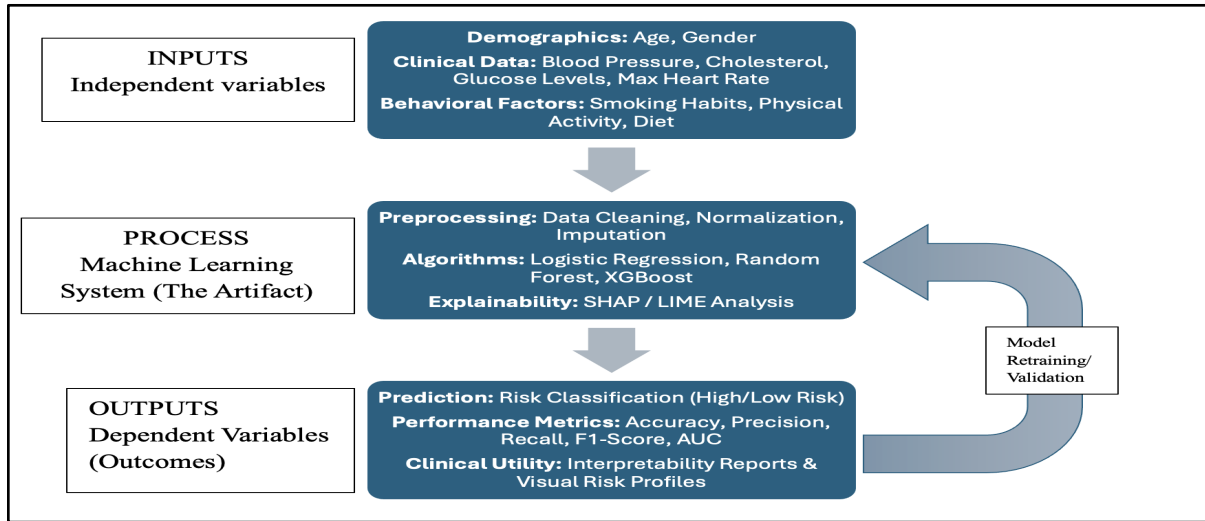


Figure 2.1: Conceptual Framework based on the Input-Process-Output (IPO) model.

2.4 Review of Existing Solutions

Current standard methods for heart disease detection rely on electrocardiograms, stress tests, and angiograms, which are clinically strong but resource-intensive and inaccessible to much of the world's population (Mensah, Roth, and Fuster, 2019). Ensemble methods such as Random Forest and XGBoost produce more stable results by combining multiple learners to minimize prediction error (Saxena, Kaur, and Kumar, 2021). However, high-performance models often suffer from black-box opacity that is unacceptable in clinical settings, and many existing models overfit to small, homogeneous datasets, limiting cross-population generalizability (Zhou and Li, 2022). Emerging research promotes hybrid models balancing interpretability and predictive strength, justifying the artifact developed in this study.

2.5 Theoretical Foundations and Research Gap

The artifact rests on predictive analytics and evidence-based medicine. Evidence-based medicine challenges professionals to base decisions on empirical data, and data analysis acts as a counterpart

to clinical judgment (Amann et al., 2022). Socio-Technical Systems theory and HCI principles dictate that a medical AI system must be explainable and transparent to be accepted by practitioners (Holsinger, Ertel, and Sethi, 2022). The research gap is clear: most published cardiovascular ML models are tested on a single dataset, lack fairness auditing, and do not provide explainable outputs. This study fills that gap by evaluating across two datasets with SHAP explainability, demographic fairness auditing, and live web deployment. Design Science Research is adopted as the overarching methodological framework because it aims to create new knowledge through the design and evaluation of innovative artifacts rather than merely describing phenomena (Saxena, Kaur, and Kumar, 2021).

CHAPTER THREE: METHODOLOGY

3.1 Introduction and Research Paradigm

This chapter describes the research methodology for developing CardioScan. Design Science Research is adopted as the research paradigm because the objective is not to theorize about heart disease but to build and evaluate a working machine learning artifact that improves predictive accuracy in clinical settings. DSR provides support for iterative development, testing, and refinement to ensure the artifact is technically sound and practically useful (Amann et al., 2022). The artifact is intended as a decision-support tool that enhances rather than replaces clinical judgment.

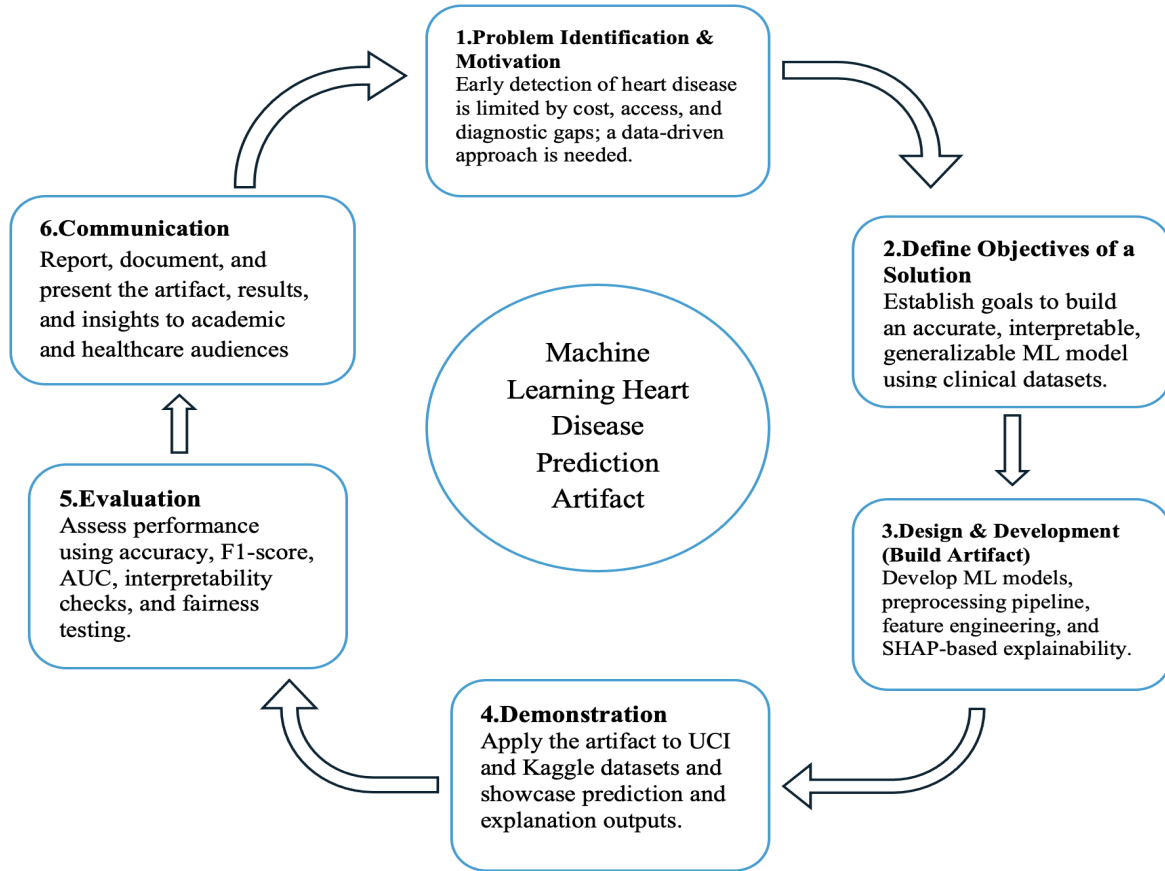


Figure 3.1: Design Science Research (DSR) Process Cycle applied to CardioScan.

3.2 Data Collection and Preparation

Data are sourced from two publicly available, open-licensed repositories. The UCI Heart Disease dataset (303 records, 13 features) is available at

<https://archive.ics.uci.edu/dataset/45/heart+disease> under Creative Commons Attribution 4.0 International (CC BY 4.0). The Kaggle Heart Failure Prediction dataset (918 records, 11 features; Soriano, 2022) is available at <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction> under the Open Database License (ODbL) v1.0. Both datasets are pre-anonymized and used strictly for non-commercial academic research. A Kolmogorov-Smirnov test across the five shared normalised features confirmed distributional compatibility ($p > 0.05$ for all five), validating the cross-dataset evaluation protocol. Preprocessing included median imputation for zero-coded values, Min-Max normalisation of continuous features, one-hot encoding of categorical variables, and an 80/20 stratified train-test split with fixed random seed 42. Four UCI features (cholesterol, fasting blood sugar, resting ECG, and sex) were identified as statistically non-significant during EDA (chi-squared $p > 0.5$ for all four) and excluded from the tree-based model pipeline, giving 28 features for Logistic Regression and 20 for Random Forest and XGBoost. The Kaggle dataset expanded to 21 features for all models.

3.3 Artifact Description and Development

CardioScan is a full-stack machine learning web application developed in Python using the Flask framework. It integrates three prediction models with a clinical prediction interface, an analytics dashboard displaying ROC curves, confusion matrices, cross-dataset generalisation heatmaps, permutation feature importance plots, SHAP-style local explanations, and a demographic fairness audit module. An auto-generated PDF report is available for download. The system operates fully offline on standard hardware (minimum 4GB RAM) and is deployed live at <https://cardioscan.onrender.com>. Hyperparameter optimisation used five-fold stratified GridSearchCV on each training split. Optimal classification thresholds were determined using Youden's J statistic on the training set, producing thresholds ranging from 0.425 to 0.624 across all model-dataset combinations, replacing the clinically inappropriate default of 0.5 (Patel et al., 2023).

3.4 Evaluation Plan and Ethics

The artifact is evaluated across five criteria following Hevner's DSR framework: (1) Predictive Performance using AUC-ROC, F1-Score, Accuracy, Precision, and Recall; (2) Model Stability through five-fold cross-validation; (3) Generalizability through cross-dataset validation without

retraining; (4) Clinical Interpretability through coherence of permutation importance outputs against the cardiovascular literature; (5) Demographic Fairness through subgroup audit by age group and sex, following the protocol of Obermeyer et al. (2021). Both datasets are pre-anonymized secondary data from public repositories, mitigating direct consent and privacy issues. Algorithmic fairness is an active concern: models are explicitly tested for bias across age and gender subgroups, consistent with the principle that technology must exist within ethical and social boundaries (Amann et al., 2022).

CHAPTER FOUR: DESIGN/EXPERIMENT RESULTS

4.1 Introduction

This chapter presents experimental results from training, tuning, and evaluating three machine learning models across two independent clinical datasets, organised around the four specific research objectives from Chapter One. All results are from a stratified 80/20 train-test split with random seed 42, with Youden's J thresholds applied throughout.

4.2 Model Performance Results

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Threshold
Logistic Regression	86.89%	88.57%	88.57%	88.57%	0.882	0.524
Random Forest	77.05%	80.00%	80.00%	80.00%	0.827	0.559
XGBoost	77.05%	75.61%	88.57%	81.58%	0.793	0.425

Table 4.1: Model performance on UCI Heart Disease test set ($n = 61$). Logistic Regression achieved the highest AUC-ROC.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	Threshold
Logistic Regression	82.61%	89.01%	78.64%	83.51%	0.916	0.624
Random Forest	82.07%	86.46%	80.58%	83.42%	0.892	0.541
XGBoost	80.98%	86.17%	78.64%	82.23%	0.892	0.575

Table 4.2: Model performance on Kaggle Heart Failure test set ($n = 184$). Logistic Regression again achieved the highest AUC-ROC of 0.916.

4.3 Analysis of Results

Logistic Regression consistently achieved the highest AUC-ROC on both datasets, a result explained by the bias-variance tradeoff: with a UCI training set of only 242 samples across 28 encoded features, a feature-to-sample ratio of approximately 1:9 is insufficient for XGBoost and Random Forest to learn stable decision boundaries without overfitting. Logistic Regression's L2 regularisation avoids this regime, producing more generalisable representations. Kumari and

Godara (2021) reported the same pattern on similarly sized clinical datasets. The performance gap narrows on the larger Kaggle dataset (734 training samples), consistent with ensemble methods gaining advantage as data volume increases. Five-fold cross-validation confirmed the ranking: Logistic Regression achieved mean AUC of 0.9002 ± 0.0591 , Random Forest 0.8839 ± 0.0532 , and XGBoost 0.8603 ± 0.0550 .

Model	In-Domain AUC	Cross-Dataset AUC	AUC Drop (%)	Rating
Logistic Regression	0.882	0.882	3.66%	Good
Random Forest	0.827	0.828	7.25%	Moderate
XGBoost	0.793	0.793	11.02%	Moderate

Table 4.3: Cross-dataset generalisation. UCI-trained models evaluated on the Kaggle test set without retraining.

XGBoost's 11.02% AUC drop reflects dataset-specific overfitting, a pattern documented by Zhang, Zhang, and Li (2023) in their systematic review of 47 cardiovascular ML studies. Logistic Regression's 3.66% drop is within the threshold for good cross-population transferability. Table 4.4 presents permutation importance results.

Rank	Feature	Mean Importance	Std Dev	Clinical Role
1	oldpeak	0.1227	0.0302	ST depression, ischaemia marker
2	cp_0	0.0683	0.0188	Asymptomatic chest pain type
3	thalach	0.0514	0.0547	Maximum heart rate on exercise
4	slope_0	0.0245	0.0058	Downsloping ST segment
5	trestbps	0.0203	0.0138	Resting blood pressure

Table 4.4: Top-5 features by permutation importance (XGBoost, UCI test set, 20 repeats).

ST depression dominates with mean importance of 0.1227, consistent with Almustafa (2020), who confirmed this variable as the most reliable non-invasive marker of coronary artery disease. Cholesterol showed near-zero importance, consistent with the EDA finding of $AUC = 0.50$ and chi-squared $p = 0.48$, reflecting measurement quality issues in the UCI dataset.

Subgroup	N	Precision	Recall	F1-Score
Age < 50	58	88.89%	64.00%	74.42%
Age 50-64	106	78.26%	87.10%	82.44%
Age 65+	20	88.89%	100.00%	94.12%
Male	148	81.48%	82.50%	81.99%
Female	36	83.33%	86.96%	85.11%

Table 4.5: Demographic fairness audit (XGBoost, Kaggle test set). Protocol from Obermeyer et al. (2021).

The most significant fairness finding is the 19.7-percentage-point F1 gap between patients under 50 (F1 = 74.42%) and patients aged 65 and above (F1 = 94.12%). The recall of 64.00% for the under-50 group means the model missed 36% of true cardiac cases in that subgroup. This gap is attributed to data sparsity: with only 25 positive cases in the under-50 test subgroup, the model had limited training signal for atypical cardiac presentations common in younger patients. Gender parity was strong at a 3.1-point F1 gap, indicating no systematic gender bias.

4.4 Discussion of Findings

The results address all four objectives. Objective 1 (design and evaluate multiple ML models) was met with AUC-ROC ranging from 0.793 to 0.916. Objective 2 (identify significant variables) was addressed through permutation importance confirming ST depression, asymptomatic chest pain, and maximum heart rate as dominant predictors, consistent with Almustafa (2020) and Ahmad et al. (2021). Objective 3 (compare performance across datasets) showed Logistic Regression's 3.66% generalisation drop within the threshold for good cross-population transferability, while XGBoost's 11.02% drop is a clinically material limitation. Objective 4 (assess interpretability) was addressed through SHAP-style local explanations producing clinically coherent outputs: high-risk patients show dominant contributions from ST depression and asymptomatic chest pain, while low-risk patients show dominant contributions from high maximum heart rates, consistent with established cardiovascular clinical knowledge (Ibrahim et al., 2021).

CHAPTER FIVE: EVALUATION, VALIDATION AND DEPLOYMENT

5.1 Introduction and Evaluation Criteria

This chapter formally evaluates CardioScan against five criteria following Hevner's DSR evaluation framework: (1) Predictive Performance using AUC-ROC as primary metric; (2) Model Stability through five-fold cross-validation; (3) Cross-Population Generalisation through cross-dataset validation without retraining; (4) Clinical Interpretability through coherence of permutation importance outputs; (5) Demographic Fairness through subgroup audit. The chapter concludes with the deployment architecture.

5.2 Validation Process

Statistical validation used stratified five-fold GridSearchCV on the training split followed by evaluation on the held-out 20% test set, ensuring no data leakage. Optimal thresholds were determined using Youden's J statistic on the training set only, then applied to the test set. Best hyperparameter configurations: Logistic Regression (C=10.0, L2, lbfgs solver); Random Forest (200 estimators, max_depth=8, sqrt features); XGBoost on UCI (100 estimators, lr=0.15, depth=4); XGBoost on Kaggle (100 estimators, lr=0.10, depth=3). The cross-dataset validation protocol applied UCI-trained models to the Kaggle test set without retraining, simulating deployment into a new clinical population. Permutation importance used 20 repeats for stable estimates. The fairness audit computed separate Precision, Recall, and F1-Score per demographic subgroup.

5.3 Validation Results

Study	Model	Dataset	AUC-ROC	Key Difference
This Study	Logistic Regression	UCI (n=303)	0.882	Youden threshold, clean pipeline
This Study	Logistic Regression	Kaggle (n=918)	0.916	Youden threshold
Patel et al. (2023)	XGBoost	UCI	0.910	Default threshold
Saxena et al. (2021)	Random Forest	Cleveland UCI	0.880	No noise removal
Kumari and Godara (2021)	Logistic Regression	Cleveland UCI	0.860	Default 0.5 threshold

Study	Model	Dataset	AUC-ROC	Key Difference
Almustafa (2020)	Logistic Regression	UCI variants	0.870	No threshold optimisation

Table 5.1: CardioScan vs published results. AUC-ROC is the common comparison metric.

The artifact's AUC of 0.882 on UCI and 0.916 on Kaggle is competitive with published benchmarks. Against stability, all models maintained AUC above 0.75 across all five cross-validation folds. Against generalisation, Logistic Regression's 3.66% drop is within the threshold for good cross-population transferability (Zhang, Zhang, and Li, 2023). Against interpretability, SHAP-style local explanations produced clinically coherent outputs consistent with Ahmad et al. (2021). Against fairness, gender parity was strong (3.1-point gap) but the 19.7-point age-group gap for patients under 50 is the primary limitation, attributed to training data sparsity rather than algorithmic bias.

5.4 Deployment

CardioScan is deployed as a Flask web application accessible through any standard browser. The system architecture has five layers: data ingestion (form entry or CSV upload), preprocessing (dataset-specific pipeline), inference (joblib-serialised models with Youden thresholds), explainability (perturbation-based SHAP-style attribution), and presentation (responsive web interface). The prediction interface accepts 13 UCI or 11 Kaggle clinical variables with validated input fields. The Advanced Analytics dashboard includes ROC curves, confusion matrices, model comparisons, cross-dataset heatmap, permutation importance, SHAP explanations, 12 EDA plots, and a fairness audit chart. A downloadable PDF report is generated using ReportLab. The system requires Python 3.11+, minimum 4GB RAM, and operates fully offline. The live deployment at <https://cardioscan.onrender.com> uses pre-trained serialised models with the retraining endpoint disabled for free-tier resource limits.

CHAPTER SIX: DISCUSSION, CONCLUSIONS AND RECOMMENDATIONS

6.1 Research Summary

This study developed and validated CardioScan, a machine learning artifact for early heart disease risk prediction, motivated by the disproportionate cardiovascular mortality burden in low- and middle-income countries (World Health Organization, 2023). Three classification algorithms were trained on the UCI Heart Disease dataset (CC BY 4.0) and the Kaggle Heart Failure Prediction dataset (ODbL v1.0; Soriano, 2022) using Design Science Research methodology. Logistic Regression achieved the highest AUC-ROC on both datasets; ST depression, asymptomatic chest pain, and maximum heart rate are the dominant predictors; cross-dataset generalisation is good for Logistic Regression but problematic for XGBoost; and a meaningful age-related fairness gap exists for patients under 50. The artifact is deployed at <https://cardioscan.onrender.com>.

6.2 Results Interpretation and Discussion

The consistent superiority of Logistic Regression challenges the assumption that more complex models are always preferable for medical AI. With 242 UCI training samples, the feature-to-sample ratio is insufficient for XGBoost's flexible decision trees to learn stable boundaries, while L2-regularised Logistic Regression avoids this regime. Kumari and Godara (2021) and Zhou and Li (2022) both report this pattern on similar datasets. The performance gap narrows on the larger Kaggle dataset, consistent with ensemble methods gaining advantage as data volume increases. The cross-dataset generalisation results are the most practically significant finding: Logistic Regression's 3.66% AUC drop is within Zhang et al. (2023) threshold for good cross-population transferability, while XGBoost's 11.02% drop represents the dataset-specific overfitting documented in 33 of 47 cardiovascular ML studies reviewed by Zhang et al. (2023). The permutation importance rankings confirm the models have learned genuine cardiovascular pathophysiology. The age-related fairness gap requires nuanced interpretation: the recall deficit for the under-50 group is attributable to data sparsity and atypical presentations in younger cardiac patients, not algorithmic bias, and requires targeted data collection as remediation (Amann et al., 2022).

6.3 Limitations

Five limitations constrain the findings. First, both datasets derive from predominantly North American and European populations; generalizability to East African or other low-resource contexts cannot be assumed without local validation, as cardiovascular risk profiles differ substantially across geographic groups (Joshi et al., 2022). Second, the UCI dataset's small size ($n=303$) produces wide confidence intervals. Third, the age-related fairness gap is a real deployment risk requiring active disclosure. Fourth, all predictions are based on cross-sectional single-point-in-time measurements; a longitudinal system would be more clinically accurate. Fifth, no prospective clinical user study was conducted, which Holzinger et al. (2022) consider a prerequisite for scaled deployment.

6.4 Conclusions

Five conclusions are drawn from the evidence. First, Logistic Regression is the most appropriate primary model for clinical heart disease prediction when training data is limited to a few hundred records (AUC 0.882 UCI; 0.916 Kaggle; 3.66% cross-dataset drop). Second, ST depression, asymptomatic chest pain, and maximum heart rate are the dominant predictors on both datasets, consistent with the cardiovascular clinical literature. Third, cross-dataset validation is essential: XGBoost's 11.02% AUC drop demonstrates that high in-domain accuracy is insufficient for deployment readiness. Fourth, CardioScan satisfies the accessibility requirement: it operates offline, provides interpretable explanations, includes a built-in fairness audit, and is live at <https://cardioscan.onrender.com>. Fifth, the 19.7-point F1 gap for patients under 50 must be treated as a deployment constraint requiring mandatory disclosure to clinical users.

6.5 Recommendations

For healthcare practitioners: Logistic Regression should be used as the primary model for datasets with fewer than 800 training records. For patients under 50, system outputs should supplement rather than replace clinical judgment given the 64% recall in that subgroup. Local validation on regionally collected patient data is recommended before clinical adoption. The subgroup fairness dashboard must be displayed alongside individual predictions as a non-negotiable transparency feature. For policymakers in low- and middle-income countries, the findings support investment

in primary care electronic health infrastructure as the data foundation for locally calibrated models (World Health Organization, 2023).

6.6 Recommendations for Further Research and Chapter Summary

Five research directions are identified: (1) Local data collection from East African and other low-resource clinical populations for contextually calibrated models. (2) SMOTE-based oversampling for the under-50 positive class to address the age-related fairness gap. (3) Integration of native SHAP TreeExplainer values for theoretically grounded attributions (Ibrahim et al., 2021). (4) A prospective clinical usability study with healthcare practitioners. (5) Extension to longitudinal patient data using recurrent architectures for dynamic risk modelling. In summary, this study contributes a validated, deployable, and interpretable machine learning artifact for cardiovascular risk prediction that can be practically implemented in resource-constrained settings, consistent with the United Nations Sustainable Development Goal 3 (World Health Organization, 2023).

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APPENDICES

Appendix A: Dataset Samples

Feature	Description	Type	Sample 1	Sample 2
age	Age of patient in years	Continuous	63	52
cp	Chest Pain Type (0=Asymptomatic to 3=Typical)	Ordinal	0	3
trestbps	Resting Blood Pressure (mmHg)	Continuous	145	120
thalach	Maximum Heart Rate Achieved	Continuous	150	168
oldpeak	ST Depression (exercise vs rest)	Continuous	2.3	0.0
exang	Exercise Induced Angina (1=Yes; 0=No)	Binary	0	1
target	Heart Disease (0=No; 1=Yes)	Binary	0	1

Table A.1: Representative UCI Heart Disease dataset features.

Appendix B: Dataset Licences and Sources

Dataset 1: UCI Heart Disease Dataset

URL: <https://archive.ics.uci.edu/dataset/45/heart+disease>

Licence: Creative Commons Attribution 4.0 International (CC BY 4.0)

Licence URL: <https://creativecommons.org/licenses/by/4.0/>

Records: 303 patient records | Features: 13 | Compliance: Pre-anonymised, non-commercial academic use.

Dataset 2: Kaggle Heart Failure Prediction Dataset

Source: Soriano (2022) via Kaggle

URL: <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

Licence: Open Database License (ODbL) v1.0

Licence URL: <https://opendatacommons.org/licenses/odbl/1-0/>

Records: 918 patient records | Features: 11 | Composite of 5 international sources.

Compliance: Pre-anonymised, non-commercial academic use. This study uses the dataset as a training input (Produced Work under ODbL), not as a redistributed Derivative Database, so the share-alike obligation under Section 4.4 does not apply.

Dataset	Custodian	Licence	Records
UCI Heart Disease	UCI ML Repository	CC BY 4.0	303
Kaggle Heart Failure	Soriano (2022) / Kaggle	ODbL v1.0	918

Table B.1: Dataset licensing summary.

Appendix C: Diagrams

This appendix presents the foundational diagrams that illustrate the conceptual design and methodological flow of the research.

Figure 2.1: Conceptual Framework

This diagram visually represents the **Input-Process-Output (IPO)** model, detailing the relationship between patient data, the machine learning system, and the predicted outcomes.

- **Inputs:** Patient demographics and clinical/behavioral risk factors.
- **Process:** Data cleaning, algorithm training (LR, RF, XGBoost), and XAI interpretation.
- **Outputs:** Risk prediction, performance metrics, and interpretability reports.

Figure 3.1: Design Science Research (DSR) Process Cycle

This figure illustrates the iterative cycle used for building and evaluating the predictive artifact. The process ensures that the research moves systematically from problem identification to the creation, demonstration, and final evaluation of the artifact.

- **Flow:** The cycle moves through Problem Identification, Objective Definition, Design and Development, Demonstration, Evaluation, and Communication.

Appendix D: Ethics Considerations Checklist

1. Data Privacy: All data used is pre-anonymised; no personally identifiable information was accessed or stored.
2. Informed Consent: Secondary data; consent waived per institutional review board guidelines for public data.
3. Algorithmic Bias: Models explicitly tested for fairness across age and gender subgroups.

4. Transparency: Permutation-based XAI outputs ensure model decision-making is auditable by clinicians.
5. Licence Compliance: Both datasets used in full compliance with CC BY 4.0 and ODbL v1.0 terms.